Enrollment No._____

GANPAT UNIVERSITY

M.Sc.IT (AI&ML) - SEM-II EXAMINATION CBCS (NEW)

P82A3ML1: MACHINE LEARNING - I

April - June 2022

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. **Answer the following: (ANY SIX FROM EIGHT)** **(30)**

a What is machine learning? Explain supervised machine learning in detail.

b Explain the steps of ML framework for developing model.

c What is linear regression? Explain linear regression in detail.

d Explain the various terms of confusion matrix in brief.

e What is LASSO? Explain LASSO regularization technique in detail.

f What is research? Explain the importance of research in detail.

g Explain decision tree algorithm in detail.

h What is classification? List down the classification algorithms in ML.

2. **Answer the following: (ANY ONE)** **(06)**

a Explain the python libraries for data analysis in brief.

b What is data analytics? Explain variance and bias in brief.

3. **Answer the following: (ANY TWO)** **(09)**

a How to build logistic regression model? Explain in brief.

b Explain ROC curve and AUC in detail.

c What is chart? Explain Gain chart in detail.

4. **Answer the following: (ANY ONE)** **(06)**

a Explain polynomial regression in detail with suitable example.

b What is regression? Explain multi linear regression in detail.

5. **Answer the following: (ANY TWO)** **(09)**

a What is KNN? Explain KNN in detail with proper example.

b Explain support vector machine algorithm in detail.

c What is classification problem? Explain naïve bayes in detail.

END OF PAPER

Enrollment No._____

GANPAT UNIVERSITY

M.Sc.IT(AI&ML) SEM-II EXAMINATION (CBCS)

P82A1PYT2 -PYTHON PROGRAMMING-2

April - June 2022

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

Q-1. Answer the following: (Any Six) **(30)**

I. What is use of File? Explain different file operation mode.

II. Write in detail about Regular Expressions.

III. How to create dataframe using csv, dictionary and list of tuples?

IV. How to draw bar and line graph using matplotlib? Explain using example.

V. Write down python script to insert employee records to the database.

VI. How to send a mail using SMTP in python? Explain using example.

VII. Explain in detail about Tkinter of python.

VIII. Explain XML Processing using python module in detail.

Q-2. Answer the following: (Any Two) **(06)**

I. Explain text and binary type of files in python.

II. Explain use *, +, and ? character in regular expression with example.

III. Explain search () and match () of re module with example.

Q3. Answer the following: (Any One) **(06)**

I. Explain head(), tail(), describe(), columns, and shape of pandas module with example.

II. How to draw pie and histo graph using matplotlib? Explain using example.

Q-4. Attempt following. **(09)**

A. Answer the following: (Any two) **(08)**

I. How to access MySQL database using phpmyadmin? Explain in brief.

II. Write down python script to select employee records from database.

III. Write down python script to delete employee records from the database.

B. Using which function you can call stored procedure from python script? **(01)**

Q5. Attempt following. **(09)**

A. Discuss following Widgets of TkInter using example. (Any Four) **(08)**

[A] Checkbutton [B] Listbox [C] Radiobutton

[D] Spinbox [E] Scrollbar

B. Write down any two advantages of GUI Programming? **(01)**

END OF PAPER

Seat No._____

GANPAT UNIVERSITY
M.Sc.IT(AI&ML) SEM-II EXAMINATION

P82B4AI - ARTIFICIAL INTELLIGENCE

April - June 2022

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. Answer the following: (ANY SIX OUT OF EIGHT) **(30)**

A. Define AI. Explain agent in AI.

B. List out the various types of Agent and explain any one with suitable diagram.

C. Explain minimax algorithm of adversarial search in detail.

D. Explain A* search algorithm in detail.

E. Explain Approaches to knowledge representation in detail.

F. Explain First order predicate calculus (FOPL) with example.

G. What is expert system? Discuss the application of expert system

H. Discuss Fuzzy logic in detail.

2. [A] Answer the following: (ANY TWO) **(08)**

a) Explain the types of environment in AI.

b) Describe the various properties of the task environment with example.

c) Explain problem solving agents with toy example.

[B] Define: Rational Agent in AI. **(01)**

3. Answer the following: (ANY ONE) **(05)**

A. Explain Breadth-first search (BFS) in detail.

B. Explain Hill climbing search in detail.

4. Answer the following: (ANY TWO) **(08)**

A. Explain Knowledge-Based Agent in detail

B. Write a note on conceptual dependency of strong slot and filler structure.

C. Explain conditional probability in detail.

5. Answer the following: (ANY TWO) **(08)**

A. Explain the case studies of MYCIN.

B. Write a note on learning.

C. Discuss the component of planning expert systems in detail.

END OF PAPER

Enrollment No._____

GANPAT UNIVERSITY

M.Sc.IT (AI &ML) Sem-II (CBCS)

P82A2DAT DATA ANALYTICS USING TOOLS

Time: 3 Hours] April - June 2022 **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. Answer the following: (Any Six) **(30)**

1. Write a short note on data collection in Machine learning.
2. Explain data cleaning in brief.
3. Discuss Pivot table with example.
4. Write a short note on regression.
5. What is node? Discuss different types of nodes in Knime.
6. Write down the steps to create bar chart in Knime.
7. What is preprocessing. Discuss it with example using weka.
8. Discuss association rules with example in weka.

2. Answer the following: (Any One) **(06)**

1. Explain Knowledge Discovery Process in detail.
2. Discuss various types of data formats.

3. Answer the following: (Any Two) **(09)**

1. What is the use of charts? Explain any two excel charts in detail.
2. Explain what if analysis with example.
3. Discuss correlation in detail.

4. Answer the following: (Any One) **(06)**

1. Draw and write down steps of Classification example performed using Knime.
2. Discuss nodes which are used to handle missing value in Knime.

5. Answer the following: (Any Two) **(09)**

1. Explain any clustering example with weka.
2. Explain different types of filters in weka.
3. Write a short note on tabs display in weka explorer.

END OF PAPER